Daffodil International University

LAB REPORT

Course Title: Computer Graphics Lab
Course Code: CSE412 & CSE422

TOPICS: Implementation of 3D Transformations.

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1 Objective

- To understand 3D transformations.
- To implement translation, rotation, and scaling in 3D.
- To observe transformation effects on 3D objects.

2 Tools

- Programming Language: C++
- Graphics Library: OpenGL (GLUT)
- IDE: CodeBlocks / GCC

3 Theory

3D transformations are used to manipulate objects in three-dimensional space.

Translation: Moves an object along x, y, and z axes.

Rotation: Rotates an object around a specified axis.

Scaling: Changes the size of the object.

In this experiment, transformations are applied to a 3D cube using OpenGL transformation functions. Keyboard inputs are used to dynamically modify the cube's position, rotation, and scale.

4 Example Used Functions

- `glTranslatef()` – 3D translation
- `glRotatef()` – 3D rotation
- `glScalef()` – 3D scaling
- `gluLookAt()` – Camera positioning
- `gluPerspective()` – Perspective projection
- `glEnable(GL_DEPTH_TEST)` – Depth control

5 Program Code

```
#include <GL/glut.h>

float angle = 0;
float posX = 0, posY = 0, posZ = 0;
float scale = 1.0;

// Cube
```

```
void drawCube() {
    glutWireCube(2);
}

// Display
void display() {
    glClear(GL_COLOR_BUFFER_BIT | GL_DEPTH_BUFFER_BIT);

    glLoadIdentity();

    gluLookAt(0, 0, 10,   0, 0, 0,   0, 1, 0);

    // =====================
    // TRANSFORMATIONS
    // =====================
    glTranslatef(posX, posY, posZ);
    glRotatef(angle, 1, 1, 0);
    glScalef(scale, scale, scale);

    glColor3f(1, 1, 1);
    drawCube();

    glutSwapBuffers();
}

// Keyboard control
void keyboard(unsigned char key, int x, int y) {
    switch(key) {
        case 'a': posX -= 0.5; break;
        case 'd': posX += 0.5; break;
        case 'w': posY += 0.5; break;
        case 's': posY -= 0.5; break;
        case 'q': posZ += 0.5; break;
        case 'e': posZ -= 0.5; break;

        case '+': scale += 0.1; break;
        case '-': scale -= 0.1; break;

        case 'r': angle += 10; break;
    }

    glutPostRedisplay();
}

// Init
void init() {
    glClearColor(0, 0, 0, 1);
    glEnable(GL_DEPTH_TEST);

    glMatrixMode(GL_PROJECTION);
    gluPerspective(60, 1, 1, 100);
```

```
    glMatrixMode(GL_MODELVIEW);
}

// Main
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_DOUBLE | GLUT_RGB | GLUT_DEPTH);
    glutInitWindowSize(700, 700);
    glutCreateWindow("3D Transformations");

    init();
    glutDisplayFunc(display);
    glutKeyboardFunc(keyboard);

    glutMainLoop();
}
```

6 Keyboard Controls

- **W / S** – Move object up / down (Y-axis)
- **A / D** – Move object left / right (X-axis)
- **Q / E** – Move object forward / backward (Z-axis)
- **R** – Rotate the object
- **+ / -** – Increase / decrease scale

7 Output

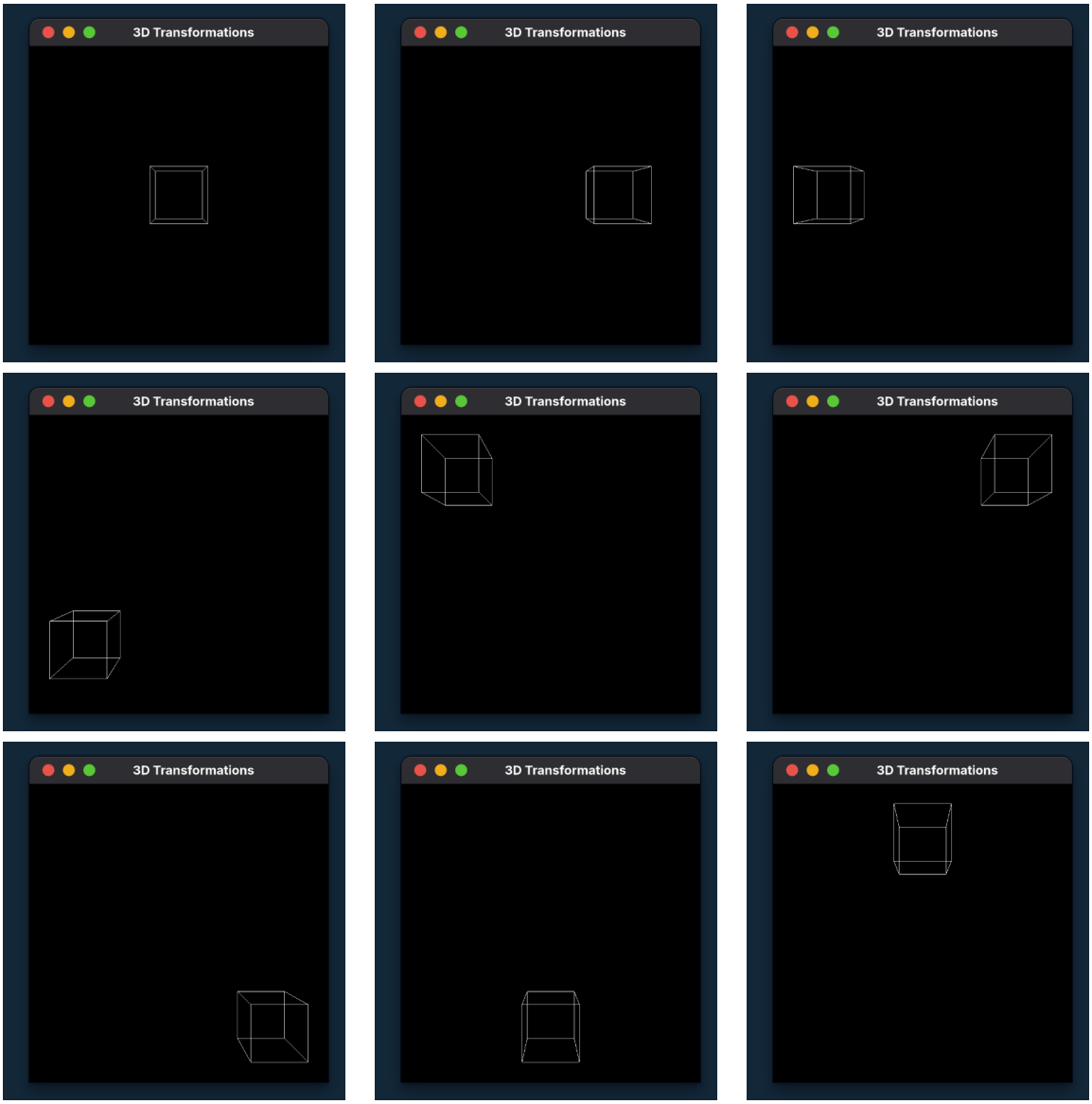

Figure 1: 3D Transformations: Translation, Rotation, and Scaling (Different States)

8 Result

The experiment was successfully completed by applying 3D transformations on a cube. The cube responded correctly to translation, rotation, and scaling operations through keyboard inputs.

It was observed that translation moved the cube along different axes, rotation changed its orientation in 3D space, and scaling adjusted its size. The combination of these transformations demonstrated how objects behave in a 3D environment.

9 Conclusion

This experiment provided a clear understanding of 3D transformations in computer graphics. It demonstrated how objects can be manipulated dynamically using translation, rotation, and scaling.

The use of keyboard controls made the transformations interactive and easy to observe. This lab strengthened the foundation for advanced 3D graphics concepts such as animation, modeling, and simulation.